

Proyecto Intermodelar 1º ASIR – Planificación y administración de Redes

Sumario

1.  Inter-Vlan Routing.....	2
2.  Tabla de direccionamiento.....	2
3.  Configuraciones.....	4
1.  Creamos las 7 vlans en los dos switch.....	4
2.  Creamos el enlace troncal entre Switches.....	4
3.  Configuramos el Router on a stick.....	5
4.  Configuración final del switch principal.....	5
5.  Configurar las subinterfaces el el router.....	5
6.  Configuramos el servidor FTP VLAN60.....	6
7.  Configurar los Puertos del switch VLAN 60.....	7
8.  Configuramos los pc's.....	7
9.  Configuración ACLs.....	7
10.  Configurar un punto de acceso.....	9
11.  Conectividad dentro de la misma VLAN.....	10
12.  Conectividad entre VLANs permitidas.....	10
13.  Bloqueo correcto por ACLs.....	11

Proyecto Intermodelar 1º ASIR – Planificación y administración de Redes

1. Inter-Vlan Routing

Implementación Inter-VLAN routing la opción elegida el Router-on-a-stick, por que es la forma más económica y sencilla para que se comuniquen entre sí, de esa manera reducimos costos al cliente.

Al usar esta opción podemos usar switch de capa 2 y un router estándar para que los datos viajen entre la red, ya que un switch de capa 3 es más caro económicamente hablando.

2. Tabla de direccionamiento

Planta	Departamento	Vlan	Red
Baja	Recepción	99	192.168.99.1
Baja	Administración	10	192.168.10.1
Baja	Dirección	20	192.168.20.1
Baja	Cpd	60	192.168.60.1
Primera	Desarrollo	30	192.168.30.1
Primera	Soporte Técnico	40	192.168.40.1
Primera	Aula de formación	50	192.168.50.1

Proyecto Intermodelar 1º ASIR – Planificación y administración de Redes

Host	Interf	Dirección IP	Mascara	Puerta enlace
PLANTA BAJA				
PC0_REC	Fa0 / Fa01	192.168.99.10	255.255.255.0	192.168.99.1
PC1_REC	Fa0 / Fa02	192.168.99.11	255.255.255.0	192.168.99.1
PC2_ADM	Fa0 / Fa03	192.168.10.10	255.255.255.0	192.168.10.1
PC3_ADM	Fa0 / Fa04	192.168.10.11	255.255.255.0	192.168.10.1
PC4_DIC	Fa0 / Fa05	192.168.20.10	255.255.255.0	192.168.20.1
PC5_DIC	Fa0 / Fa06	192.168.20.11	255.255.255.0	192.168.20.1
SERVER0	Fa0 / Fa07	192.168.60.10	255.255.255.0	192.168.60.1
SERVER1	Fa0 / Fa08	192.168.60.11	255.255.255.0	192.168.60.1
PRIMERA PLANTA				
PC6_DES	Fa0 / Fa01	192.168.30.10	255.255.255.0	192.168.30.1
PC7_DES	Fa0 / Fa02	192.168.30.11	255.255.255.0	192.168.30.1
PC8_ST	Fa0 / Fa03	192.168.40.10	255.255.255.0	192.168.40.1
PC9_ST	Fa0 / Fa04	192.168.40.11	255.255.255.0	192.168.40.1
PC10_AF	Fa0 / Fa05	192.168.50.10	255.255.255.0	192.168.50.1
PC11_AF	Fa0 / Fa06	192.168.50.11	255.255.255.0	192.168.50.1

Proyecto Intermodelar 1º ASIR – Planificación y administración de Redes

3. Configuraciones

1. Creamos las 7 vlans en los dos switch

```
enable  
configure terminal
```

```
vlan 10  
name Admin  
exit
```

```
vlan 20  
name Direc  
exit
```

```
vlan 30  
name Desa  
exit
```

```
vlan 40  
name Soporte  
exit
```

```
vlan 50  
name Formacion  
exit
```

```
vlan 60  
name CPD  
exit
```

```
valn 99  
name Recepcion  
exit
```

2. Creamos el enlace troncal entre Switches

en el puerto que conecta los dos switch (GIG02)

```
interface GigabitEthernet 0/2  
switchport mode trunk
```

Proyecto Intermodelar 1º ASIR – Planificación y administración de Redes

3. Configuramos el Router on a stick

- Entra a la interfaz conectada al switch ej GIG0/0 y la encendemos

```
interface GigabitEthernet 0/0  
no shutdown
```

- Creamos una subinterfaz por cada departamento (VLAN)

```
interface GigabitEthernet 0/0.10  
encapsulation dot1Q 10  
ip address 192.168.10.1 255.255.255.0
```

```
interface GigabitEthernet 0/0.20  
encapsulation dot1Q 20  
ip address 192.168.20.1 255.255.255.0
```

4. Configuración final del switch principal

El puerto del switch que va al router debe estar en modo trunk

1 Switch planta 1

```
interface GigabitEthernet 0/2  
switchport mode trunk
```

5. Configurar las subinterfaces el el router

para que todas las VLANs se comuniquen con el servidor FTP, el router necesita una subinterfaz por cada rec. Entramos a la interfaz conectada al switch GIG0/1

```
enable  
configure terminal  
interface GigabitEthernet 0/1  
no ip address  
no shutdown  
exit
```

Proyecto Intermodelar 1º ASIR – Planificación y administración de Redes

```
# VLAN 10 - Admin
interface GigabitEthernet 0/1.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
 exit
```

```
# VLAN 20 - Dir
interface GigabitEthernet 0/0.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
 exit
```

```
# VLAN 30 - Desa
interface GigabitEthernet 0/0.30
 encapsulation dot1Q 30
 ip address 192.168.30.1 255.255.255.0
 exit
```

```
# VLAN 40 – Soporte
interface GigabitEthernet 0/0.40
 encapsulation dot1Q 40
 ip address 192.168.40.1 255.255.255.0
 exit
```

```
# VLAN 50 - Formacion
interface GigabitEthernet 0/0.50
 encapsulation dot1Q 50
 ip address 192.168.50.1 255.255.255.0
 exit
```

```
# VLAN 60 - CPD
interface GigabitEthernet 0/0.60
 encapsulation dot1Q 60
 ip address 192.168.60.1 255.255.255.0
 exit
```

```
# VLAN 99 - Recepcion
interface GigabitEthernet 0/0.99
 encapsulation dot1Q 99
 ip address 192.168.99.1 255.255.255.0
 exit
```

6. Configuramos el servidor FTP VLAN60

Conectamos el servidor al switch planta 1 Fa0/24

Proyecto Intermodelar 1º ASIR – Planificación y administración de Redes

IP: 192.168.60.10
Mascara: 255.255.255.0
Gateway 192.168.60.1

Servicio FTP: Pestaña Services FTP, ON y crear un usuario

7. Configurar los Puertos del switch VLAN 60

Switch planta 1

```
vlan 60
name Servidores
exit
interface fastEthernet 0/24
switchport mode access
switchport access vlan 60
```

8. Configuramos los pc's

en cualquier pc
vlan 10 – Administración

IP 192.168.10.10
Gateway: 192.168.1.10.1

9. Configuración ACLs

En el router

1· Bloquear Aula VLAN50 hacia Dirección VLAN20

```
ip access-list extended ACL_AULA
deny ip 192.168.50.0 0.0.0.255 192.168.20.0 0.0.0.255
permit ip any any
exit
```

interface vlan 50 (o gi0/0.50 en Router)

Proyecto Intermodelar 1º ASIR – Planificación y administración de Redes

ip access-group ACL_AULA in

2· Bloquear Desarrollo VLAN30 hacia Administración VLAN10

```
ip access-list extended ACL_DESARROLLO
deny ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255
permit ip any any
exit
```

```
interface vlan 30 (o gi0/0.30 en Router)
ip access-group ACL_DESARROLLO in
```

3· Permitir Administración VLAN10 a Servidores VLAN60

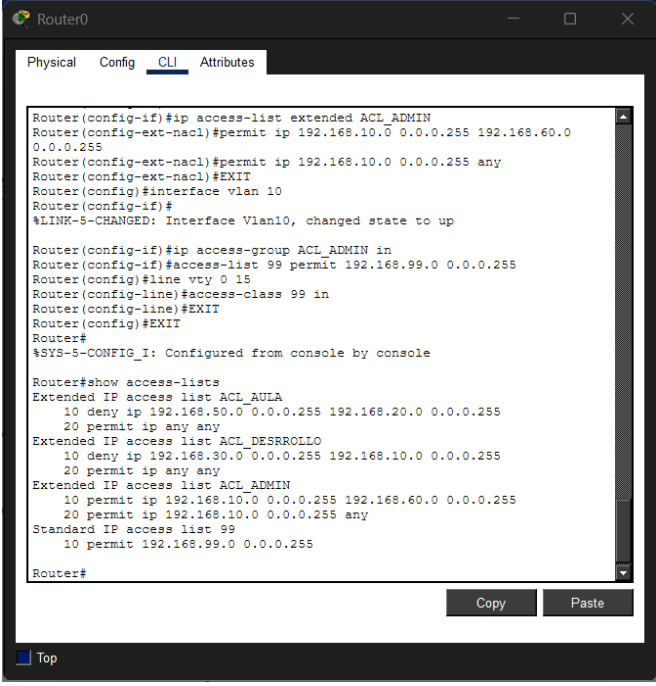
```
ip access-list extended ACL_ADMIN
permit ip 192.168.10.0 0.0.0.255 192.168.60.0 0.0.0.255
permit ip 192.168.10.0 0.0.0.255 any
exit
```

```
interface vlan 10 (o gi0/0.10 en Router)
ip access-group ACL_ADMIN in
```

4· Solo la VLAN MGMT VLAN99 administra dispositivos

```
access-list 99 permit 192.168.99.0 0.0.0.255
```

```
line vty 0 15
access-class 99 in
exit
```



```
Router0
Physical Config CLI Attributes

Router(config-if)#ip access-list extended ACL_ADMIN
Router(config-ext-nacl)#permit ip 192.168.10.0 0.0.0.255 192.168.60.0
0.0.0.255
Router(config-ext-nacl)#permit ip 192.168.10.0 0.0.0.255 any
Router(config-ext-nacl)#EXIT
Router(config)#interface vlan 10
Router(config-if)#
%LINK-5-CHANGED: Interface Vlan10, changed state to up

Router(config-if)#ip access-group ACL_ADMIN in
Router(config-if)#access-list 99 permit 192.168.99.0 0.0.0.255
Router(config)#line vty 0 15
Router(config-line)#access-class 99 in
Router(config-line)#EXIT
Router(config)#EXIT
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show access-lists
Extended IP access list ACL_AULA
 10 deny ip 192.168.30.0 0.0.0.255 192.168.20.0 0.0.0.255
 20 permit ip any any
Extended IP access list ACL_DESARROLLO
 10 deny ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255
 20 permit ip any any
Extended IP access list ACL_ADMIN
 10 permit ip 192.168.10.0 0.0.0.255 192.168.60.0 0.0.0.255
 20 permit ip 192.168.10.0 0.0.0.255 any
Standard IP access list 99
 10 permit 192.168.99.0 0.0.0.255

Router#
```


Proyecto Intermodelar 1º ASIR – Planificación y administración de Redes

10. Configurar un punto de acceso

preparar las VLANs en el Switch

```
interface FastEthernet0/10 # Puerto donde está el AP
switchport mode trunk
switchport trunk allowed vlan 10,50
```

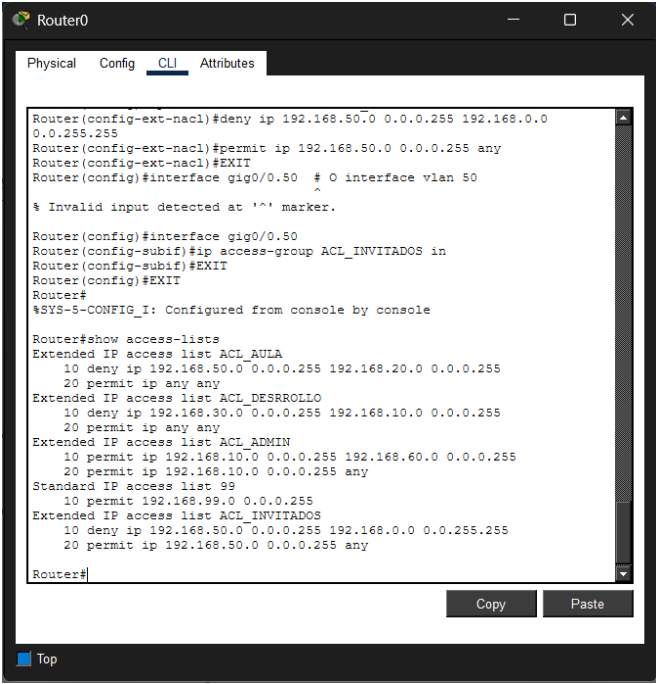
Configurar el punto de acceso

SSID: SSID_ESTANCO
Authentication: WPA2-PSK
VLAN ID:10

En el router para aislar a los invitados y que no vean la empresa:

```
# Definir la ACL para Invitados
ip access-list extended ACL_INVITADOS
# Denegar acceso a todas las redes internas (192.168.0.0/16 cubre todas tus VLANs)
deny ip 192.168.50.0 0.0.0.255 192.168.0.0 0.0.255.255
# Permitir todo lo demás (Internet)
permit ip 192.168.50.0 0.0.0.255 any
exit

# Aplicar a la interfaz de la VLAN de invitados
interface gig0/0.50 # O interface vlan 50
ip access-group ACL_INVITADOS in
```

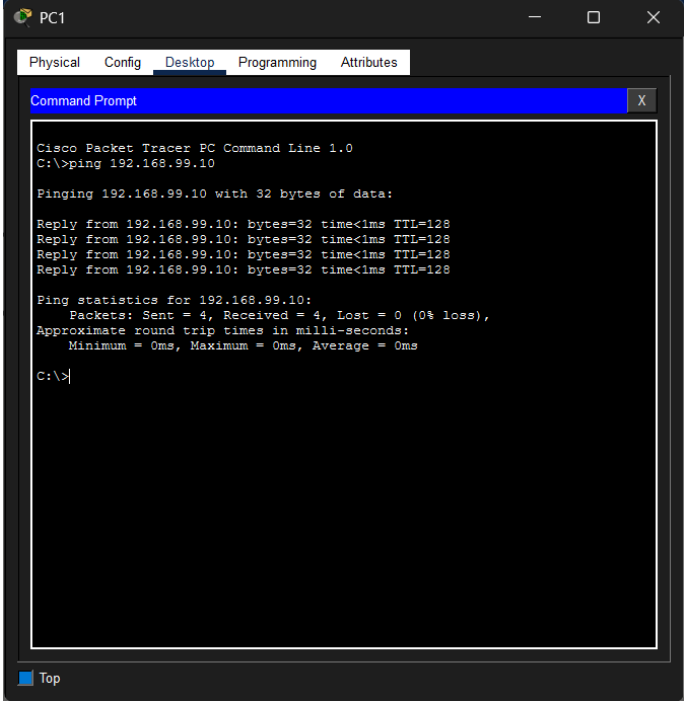


```
Router0
Physical Config CLI Attributes
Router(config-ext-nacl)#deny ip 192.168.50.0 0.0.0.255 192.168.0.0
0.0.255.255
Router(config-ext-nacl)#permit ip 192.168.50.0 0.0.0.255 any
Router(config-ext-nacl)#EXIT
Router(config)#interface gig0/0.50 # O interface vlan 50
% Invalid input detected at '^' marker.
Router(config)#interface gig0/0.50
Router(config-subif)#ip access-group ACL_INVITADOS in
Router(config-subif)#EXIT
Router(config)#EXIT
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show access-lists
Extended IP access list ACL_AULA
 10 deny ip 192.168.50.0 0.0.0.255 192.168.20.0 0.0.0.255
 20 permit ip any any
Extended IP access list ACL_DESRROLLO
 10 deny ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255
 20 permit ip any any
Extended IP access list ACL_ADMIN
 10 permit ip 192.168.10.0 0.0.0.255 192.168.60.0 0.0.0.255
 20 permit ip 192.168.10.0 0.0.0.255 any
Standard IP access list 99
 10 permit 192.168.99.0 0.0.0.255
Extended IP access list ACL_INVITADOS
 10 deny ip 192.168.50.0 0.0.0.255 192.168.0.0 0.0.255.255
 20 permit ip 192.168.50.0 0.0.0.255 any
Router#
```

Proyecto Intermodelar 1º ASIR – Planificación y administración de Redes

11. 🤝 Conectividad dentro de la misma VLAN



The screenshot shows the Cisco Packet Tracer PC Command Line interface for PC1. The 'Desktop' tab is selected, and the 'Command Prompt' window is open. The user has entered the command 'ping 192.168.99.10'. The output shows four successful replies from 192.168.99.10 with 32 bytes of data, a TTL of 128, and a time of 0ms. The ping statistics for 192.168.99.10 show 4 packets sent, 4 received, 0 lost (0% loss), and approximate round trip times of 0ms.

```
PC1
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.99.10

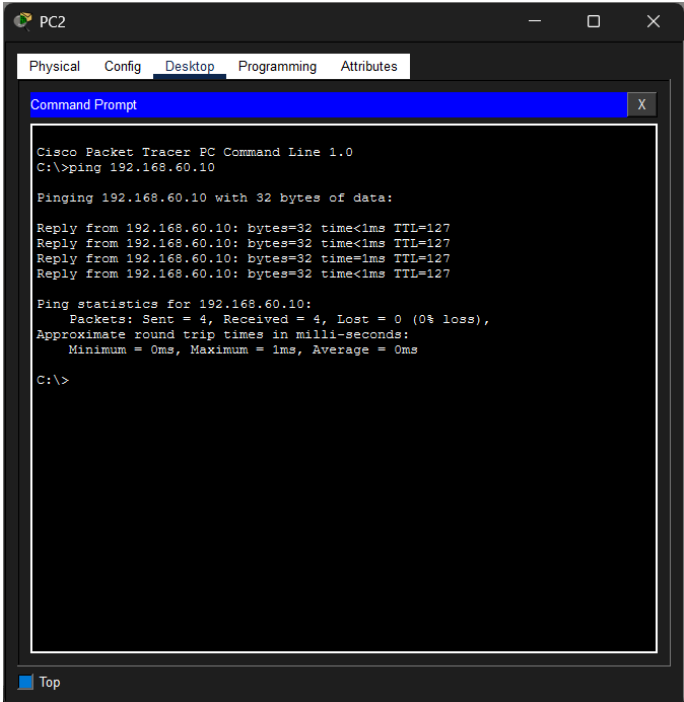
Pinging 192.168.99.10 with 32 bytes of data:

Reply from 192.168.99.10: bytes=32 time<1ms TTL=128
Reply from 192.168.99.10: bytes=32 time<1ms TTL=128
Reply from 192.168.99.10: bytes=32 time<1ms TTL=128
Reply from 192.168.99.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.99.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

12. ⚡ Conectividad entre VLANs permitidas



The screenshot shows the Cisco Packet Tracer PC Command Line interface for PC2. The 'Desktop' tab is selected, and the 'Command Prompt' window is open. The user has entered the command 'ping 192.168.60.10'. The output shows four successful replies from 192.168.60.10 with 32 bytes of data, a TTL of 127, and a time of 1ms. The ping statistics for 192.168.60.10 show 4 packets sent, 4 received, 0 lost (0% loss), and approximate round trip times of 1ms.

```
PC2
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.60.10

Pinging 192.168.60.10 with 32 bytes of data:

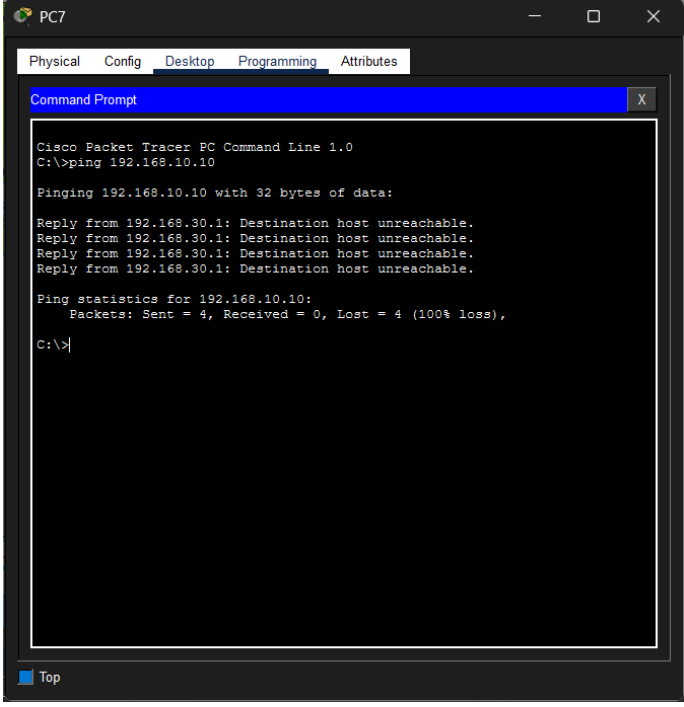
Reply from 192.168.60.10: bytes=32 time<1ms TTL=127
Reply from 192.168.60.10: bytes=32 time<1ms TTL=127
Reply from 192.168.60.10: bytes=32 time<1ms TTL=127
Reply from 192.168.60.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.60.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Proyecto Intermodelar 1º ASIR – Planificación y administración de Redes

13. Bloqueo correcto por ACLs



The screenshot shows a Cisco Packet Tracer PC Command Line window for PC7. The window has tabs for Physical, Config, Desktop, Programming, and Attributes. The Command Prompt is open, showing the output of a ping command to 192.168.10.10. The output indicates that the destination host is unreachable for all four attempts, resulting in a 100% loss of packets.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```